## Cold Weather Operations Kit (S.O.O. 6276 or S.O.O. 6287) – General data

1. General

The Cold Weather Operations Kit 115 VAC (S.O.O. 6287) or 230 VAC (S.O.O. 6276) is used to heat the engines, engine oil, Primary Flight Displays (PFDs), and the batteries. The system reduces thermal stress, warm-up times, and increases the reliability of safety operations in cold environments.

2. Description

Cold Weather Operations are facilitated by incorporating a series of heating elements onto engines, oil reservoirs, PFDs and batteries for use when the aircraft is on ground. Power is provided through a power source and shore power line supplied by the operator to the system, which is separated from, and independent of aircraft systems. Additionally, provisions are provided for an outlet to power a portable heater inside the cockpit. The system is not operational during flight.

An option for either 115 VAC (S.O.O. 6287) or 230 VAC (S.O.O. 6276) is provided depending on the requirements of the operator. For 230 VAC operations, adaptor plugs are provided to replace the ends of power cords used for the system. Power cords are to be supplied by the operators.

System heat is provided using electrical resistance heating elements in the form of thin pads, sized for the various locations. Heating elements are bonded to the engine, below the PFDs and to the bottom of the emergency lighting battery (when equipped), with a high temperature fluorosilicone adhesive. The main battery is heated using a battery blanket wrapped around the battery. Both battery and PFD elements are controlled using a thermostatic switch set to turn on heating below 32°F (0°C) and turn off heating once temperatures have risen back above 32°F (0°C), when the system is energized. All other elements are actively heating whenever the system is energized.

Power is routed to the elements through a dedicated wiring harness, routed alongside aircraft power wiring and stood off using nylon d-blocks. Power enters the system through a dedicated ground power port located in the hydraulic bay access door. System activation is identified by LED lights mounted above the door for easy visibility.

3. Materials and Equipment

A. Materials

- DC 730, Dow Corning High temperature fluorosilicone adhesive/sealant
- Industrial solvent
  (For cleaning gluing surfaces prior to application of fresh adhesive, refer to 20-40-01, Silicone Adhesives – General Data, Table 1 – Solvent Selection for General Materials, or MEK, or CB911, Click Bond™ Solvent Wipes)
  (For removing old adhesive use Acetone or DOW CORNING® DS–2025 Silicone Cleaning Solvent)
- Securing Material/Fixture, for element placement during cure
- Optional primer: DOW CORNING® DC 1200 Prime Coat, adhesion promoter.
  (Not required when using the CB911 Solvent Wipe, though suggested in rough service application)
- Abrasive cleaning pad (For example, Scotch Brite™, green or maroon)

B. Equipment

- Clean, lint free rags or cloth conforming to AMS 3819
- Plastic Scraper (putty knife)
- Calibrated multi-meter

4. Requirements

A. Element Location

1) The installation site is to be smooth and clear of obstructions.

2) The mounting site and the area immediately adjacent to the pad element perimeter must be clean in order to allow for proper sealant and tape adhesion, and/or fixture attachment.

3) Pre-fitting trial installation without sealant is suggested for all installations. This should include the fitting of fixtures for first time and complex installations.

4) Consider wire lead orientation, system installation instruction, and cable routing.

5) Tape or custom formed fixture(s) must be capable of maintaining pressure on the pad element throughout the cure process.

B. Adhesive/Sealant Cure Process

**WARNING** **DO NOT TOUCH ANY OF THE ELEMENTS WHILE THE SYSTEM IS IN OPERATION AS ELEMENTS MAY BURN EXPOSED SKIN.**

**CAUTION** **DO NOT ATTEMPT TO CURE THE BONDING AGENT BY APPLYING POWER TO THE ELEMENT OR THE PREHEAT SYSTEM.**

1) A minimum undisturbed full cure time of 8 hours is recommended in standard conditions of 21°C (70°F) with humidity level of 50% before the removal of tape, installation blocks, or other securing devices.

2) Cure time is dependant on a number of variables such as thickness of the bonding agent, temperature, and humidity levels. Due to the many variables under standard conditions it is possible to return to service with a minimum undisturbed initial cure time of 2 hours with the completion of the final cure process under engine operating temperatures.

3) If the short initial cure is used, removal of securing materials must be done in a manner that avoids disturbing the heat element and sealant.

4) The processing temperature (cure temp) for DC 730 is between 0°C (32°F) and 80°C (175°F). The operating temperature of the aircraft engine can complete the cure of the adhesive/sealant as long as the above value 80°C (175°F) is not exceeded.

5. Surface Preparation

Surfaces can vary greatly. This can affect sealant adhesion. It is recommended that adhesion characteristics to a specific surface be determined prior to application on parts or assemblies.

A. Cleaning

1) A progressive cleaning procedure should be employed using appropriate solvents (refer to Para 3.A) and a new lint free cloth conforming to AMS 3819. Contaminants such as dirt, grease, and/or processing lubricants must be removed prior to adhesive/sealant application. Always pour solvent on the cloth to avoid contaminating the solvent supply. Wash one small area at a time.

a) Clean surface with solvent, using rolling motion to remove surface contamination. Inspect cloth and repeat if contamination on wiping cloth is visible. Repeat using clean solvent wipe until solvent wiping cloth remains clean upon inspection.

b) Immediately after solvent wipe, use second clean dry cloth and wipe surface using rolling motion to remove residual solvent prior to the solvent evaporating. This is done to prevent redepositing contaminants on the surface. Do not allow solvent to evaporate.

c) Once surface is clean, abrade using a non-metallic abrasive cleaning pad and solvent.

d) After abrading, outline where the pad is to be installed.

e) Aggressively wipe down the installation site and the bonding side of pad element, with the solvent, refer to Para 3.A, followed immediately by dry wipe.

6. Removal/Installation

Refer to Figure 1 and Figure 2 for details.

A. Removal of Heating Pads

**CAUTION** **DO NOT USE ANY OTHER BONDING AGENTS, SILICONE OR RTV. TO AVOID DAMAGE TO THE SURFACE OR FINISH OF THE INSTALLATION SITE, USE APPROPRIATE SOLVENT. REFER TO Para 3.A FOR SOLVENT SELECTION.**

1) Engine Heating Pads

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Gain access to the engines by opening the lower cowls.

c) Locate the heating pad and remove it with a non-marring scraper that will not damage the surface or the finish that the heating pad is bonded to.

d) Use the scraper to lift and release a corner of the heating pad. Grip the released corner firmly, and slowly pull the heating pad away from the mounted surface.

e) Use the scraper to separate the heating pad from the surface where required.

f) Use of a solvent during the removal process may ease removal of the pad. Refer to Para 3.A.

g) Disconnect the heating pad's wire from the junction block. Pay special attention to the wire routing and attachment details for heating pad reinstallation.

h) Clean the surface with appropriate solvent and scraper. Refer to Para 3.A.

Note It is not recommended that removed elements be reused.

2) Emergency Battery Heating Pad

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Gain access to the battery bay by removing the floor panel. (Refer to 12-00-00, Servicing – General Data, Figure 2 – Servicing Diagram).

c) Disconnect the heating pad's connector from the electrical system. Pay special attention to the wire routing and attachment details for heating pad reinstallation.

d) Remove the battery. (Refer to 33-50-01, Emergency Lighting Battery – General Data).

e) Locate and remove the heating pad with non-marring scraper that will not damage the surface or the finish that the heating pad is bonded to.

f) Use the scraper to lift and release a corner of the heating pad.

g) Grip the released corner firmly, and slowly pull the heating pad away from the mounted surface.

h) Use the scraper to separate the heating pad from the surface where required.

i) Use of solvent during the removal process may ease removal of the pad. Refer to Para 3.A.

j) Clean the surface with an appropriate solvent and tool.

k) Reinstall the battery. (Refer to 33-50-01, Emergency Lighting Battery – General Data).

Note It is not recommended that removed elements be reused.

3) Main Battery Heating Belt

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Gain access to the battery bay by removing the floor panel. (Refer to 12-00-00, Servicing – General Data, Figure 2 – Servicing Diagram).

c) Disconnect the heating belt's connector from the electrical system. Pay special attention to the wire routing and attachment details for heating pad reinstallation.

d) Remove the battery. (Refer to 24-30-31, Main Battery System – Maintenance Practices).

e) Cut the cable ties that secure the battery belt to the battery.

f) Remove the battery heating belt.

g) Reinstall the battery. (Refer to 24-30-31, Main Battery System – Maintenance Practices).

4) Primary Flight Display Heating Pads

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Remove the pilot and co-pilot's PFDs, refer to 31-11-00, Primary Flight Display (PFD), Para 4.

c) Disconnect the heating pad's connector from the electrical system. Pay special attention to the wire routing and attachment details for heating pad reinstallation.

d) Locate the heating pad and remove it with a non-marring scraper that will not damage the surface or the finish that the heating pad is bonded to.

e) Use the scraper to lift and release a corner of the heating pad.

f) Grip the released corner firmly, and slowly pull the heating pad away from the mounted surface.

g) Use the scraper to separate the heating pad from the surface where required.

h) Use of solvent during the removal process may ease removal of the pad. Refer to Para 3.A.

i) Clean the surface with an appropriate solvent and tool.

Note It is not recommended that removed elements be reused.

B. Installation of Heating Pads

**CAUTION DO NOT USE ANY OTHER BONDING AGENTS, SILICONE OR RTV. TO AVOID DAMAGE TO THE SURFACE OR FINISH OF THE INSTALLATION SITE, USE APPROPRIATE SOLVENT. REFER TO Para 3.A FOR SOLVENT SELECTION.**

**CAUTION DO NOT COAT THE TOP SURFACE OF THE PAD ELEMENT WITH ADHESIVE/SEALANT.**

1) Engine Heating Pads

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Gain access to both engines by opening the lower cowls.

c) Locate and clean each installation site with an appropriate solvent and tool as per Para 5.A. Refer to Para 3.A.

d) Refer to Table 1 — Heating Pad Lead Direction for lead positioning.

e) Confirm resistance values. Refer to Table 2 — 115V System Values (±7%) and Table 3 — 230V System Values (±7%).

f) Spread DC730 Flourosilicone adhesive into a thin layer between 0.020 - 0.032 inches thick to the heating pad contact surface (flat, unmarked side).

Note The heating pad must be in complete bonded contact with the mounting surface through the adhesive/sealant. Cavities in the installation site directly under the heating pad must be filled with adhesive/sealant in order to provide proper thermal transfer.

g) Place the heating pad in location and bond, using a roller or plastic putty applicator.

h) Starting at the center of the pad, draw the roller across the surface of the pad to the outer edge, expelling air bubbles, cavities, and imperfections. When the proper amount of sealant is used, very little sealant will be expelled from under the pad.

i) When cleaning excess sealant after rolling, use extra care, and sealant, as required to seal the edges of the pad.

j) Use tooling (vacuum bags, tape, foam, straps, etc.) to ensure that constant and even pressure is applied to the pad during curing.

k) Once the bonding agent has cured (refer to Para 4.B) remove bag, tape and/or fixtures.

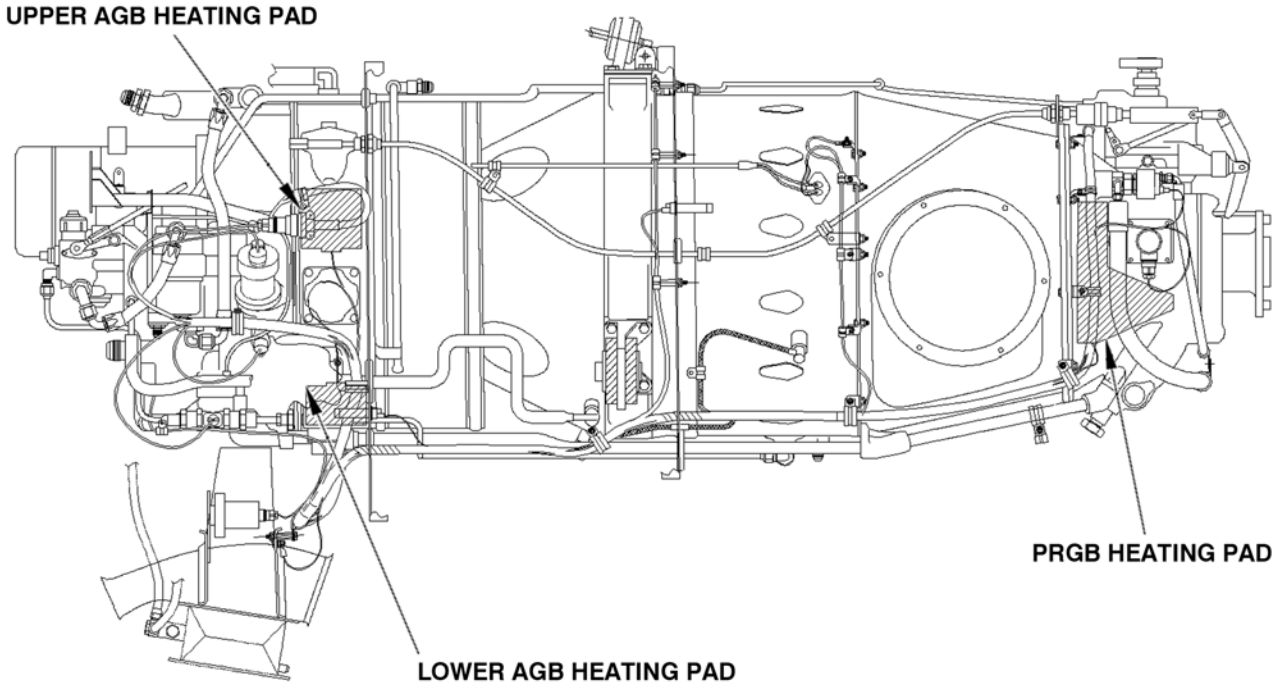
l) Apply a dollop of the adhesive/sealant under the base of the element lead (where lead enters element). This reduces wire fatigue and guards against the incursion of fluids at this location.

m) Seal the perimeter of the element with adhesive/sealant. Use just enough of the sealant to create a smooth transition along the edge of the element to mounting surface. This transition is not to extend beyond one quarter of an inch (.25”) from the element and not more than one eighth of an inch (0.125”) onto the elements upper surface.

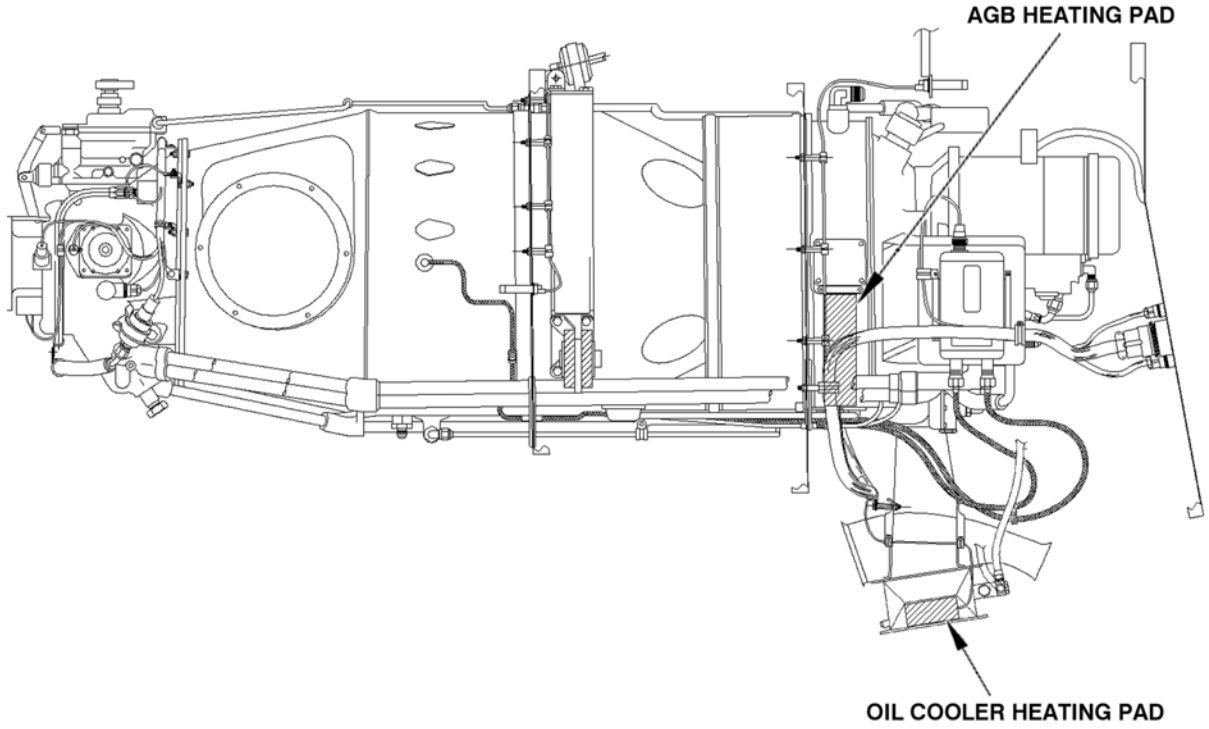
n) Refer to PSM 1-64-2W, 25-70-00, Cold Weather Operations Kit (S.O.O.s 6287 or 6276) for wire routing and terminations. If reinstalling a heating pad, route wires to match those previously removed in Para 6.A.1).

**Table 1 Heating Pad Lead Direction**

| Heating Pad | Lead Direction |
|---|---|
| Left Side AGB | Position heating pad with the lead down |
| Right Side AGB, Upper Pad | Position heating pad with the lead down |
| Right Side AGB, Lower Pad | Position heating pad with the lead up |
| Right Side PRGB | Position heating pad with the lead down |
| Left Side of Engine Oil Cooler | Position heating pad with the lead aft |




Right Hand Side of Engine
Figure 1




Left Hand Side of Engine
Figure 2

2) Emergency Lighting Battery Heating Pad (If Installed)

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Gain access to the battery bay by removing the floor panel. (Refer to 12-00-00, Servicing – General Data, Figure 2 – Servicing Diagram).

c) Remove the battery. (Refer to 33-50-01, Emergency Lighting Battery – General Data) (if installed).

d) Clean the installation site with an appropriate solvent and non-marring scraper as per Para 5.A. Refer to Para 3.A for materials and equipment.

e) Confirm resistance values. Refer to Table 2 — 115V System Values (±7%) and Table 3 — 230V System Values (±7%).

f) Position the lead in the direction of existing wiring, refer to Figure 3.

g) Spread DC730 Flourosilicone adhesive into a thin layer between 0.020 - 0.032 inches thick to the heating pad contact surface (flat, unmarked side).

Note The heating pad must be in complete bonded contact with the mounting surface through the adhesive/sealant. Cavities in the installation site directly under the heating pad must be filled with adhesive/sealant in order to provide proper thermal transfer.

h) Place the heating pad in location and bond, using a roller or plastic putty applicator.

i) Starting at the center of the pad, draw the roller across the surface of the pad to the outer edge, expelling air bubbles, cavities, and imperfections. When the proper amount of sealant is used, very little sealant will be expelled from under the pad.

j) When cleaning excess sealant after rolling, use extra care, and sealant, as required to seal the edges of the pad.

k) Use tooling (vacuum bags, tape, foam, straps, etc.) to ensure that constant and even pressure is applied to the pad during curing.

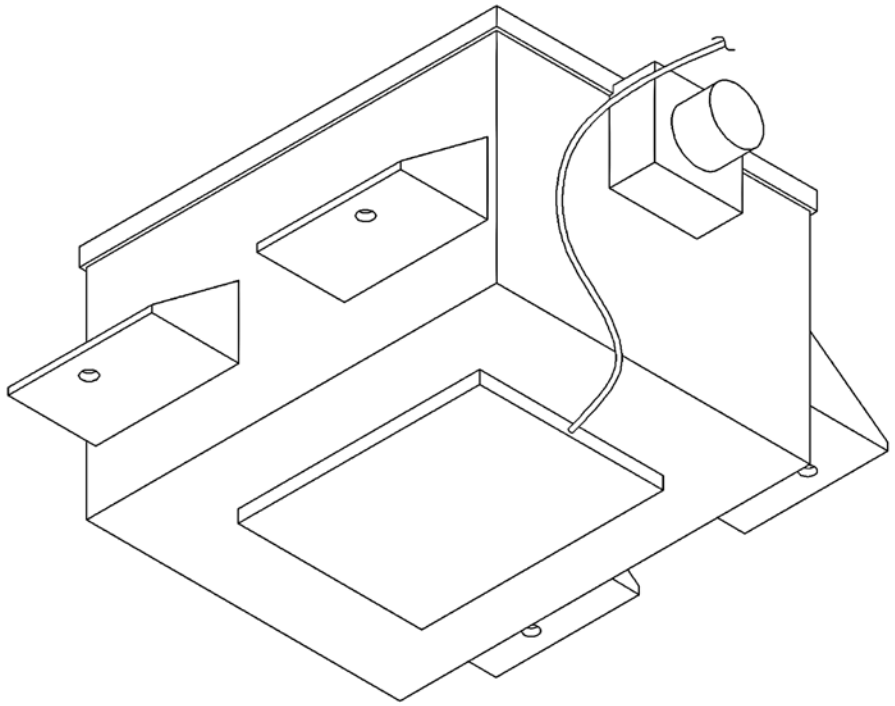
l) Once the bonding agent has cured (refer to Para 4.B) remove bag, tape and/or fixtures.

m) Apply a dollop of the adhesive/sealant under the base of the element lead (where lead enters element). This reduces wire fatigue and guards against the incursion of fluids at this location.

n) Seal the perimeter of the element with adhesive/sealant. Use just enough of the sealant to create a smooth transition along the edge of the element to mounting surface. This transition is not to extend beyond one quarter of an inch (.25”) from the element and not more than one eighth of an inch (0.125”) onto the elements upper surface.

o) Install the battery. (Refer to 33-50-01, Emergency Lighting Battery – General Data).

p) Connect the heating pad electrical connector to the appropriate mating wire harness connector. Route wires to match those previously removed in Para 6.A.2).



Emergency Lighting Battery
Figure 3

3) Main Battery Heating Belt

   a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

   b) Gain access to the battery bay by removing the floor panel. (Refer to 12-00-00, Servicing – General Data, Figure 2 – Servicing Diagram).

   c) Remove the battery. (Refer to 24-30-31, Main Battery System – Maintenance Practices).

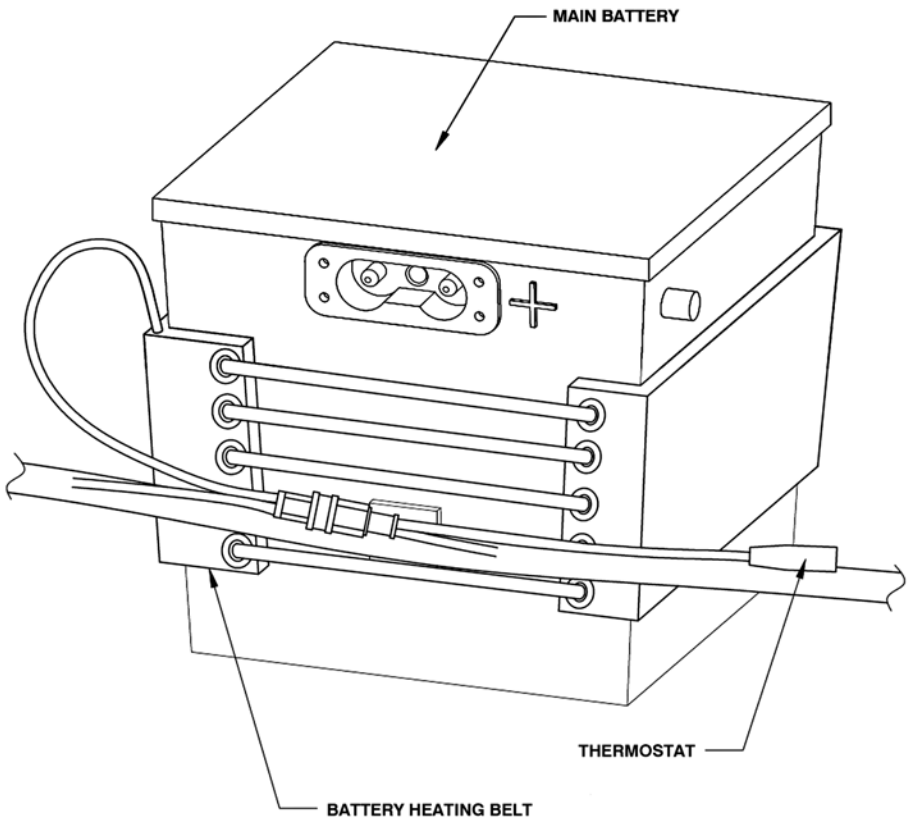
   d) Wrap the battery belt around the main battery. Position the eyelets on either side of the battery's main terminal, refer to Figure 4.

   e) Secure heating belt around the battery by looping cable ties through parallel eyelets, refer to Figure 4.

   f) Gently tighten the cable ties and alternate tension between the ties. Do not pull too hard as this could damage the belt or remove eyelets.

   g) Reinstall the battery. (Refer to 24-30-31, Main Battery System – Maintenance Practices).

   h) Connect the heating belt electrical connector to the appropriate mating wire harness connector. Route wires to match those previously removed in Para 6.A.3).




Main Battery
Figure 4

4) Primary Flight Display Heating Pads

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Remove the pilot and co-pilot's PFDs, refer to 31-11-00, Primary Flight Display (PFD), Para 4.

c) Clean the top surface of the frame below both PFDs with an appropriate solvent, refer to Para 3.A, and tool as per Para 5.A.

d) Position the heating pads with the lead forward and toward the left side, refer to Figure 5.

e) Confirm resistance values. Refer to Table 2 — 115V System Values (±7%) and Table 3 — 230V System Values (±7%).

f) Spread DC730 Flourosilicone adhesive into a thin layer between 0.020 - 0.032 inches thick to the heating pad contact surface (flat, unmarked side).

Note The heating pad must be in complete bonded contact with the mounting surface through the adhesive/sealant. Cavities in the installation site directly under the heating pad must be filled with adhesive/sealant in order to provide proper thermal transfer.

g) Place the heating pad in location and bond, using a roller or plastic putty applicator.

h) Starting at the center of the pad, draw the roller across the surface of the pad to the outer edge, expelling air bubbles, cavities, and imperfections. When the proper amount of sealant is used, very little sealant will be expelled from under the pad.

i) When cleaning excess sealant after rolling, use extra care, and sealant, as required to seal the edges of the pad.

j) Use tooling (vacuum bags, tape, foam, straps, etc.) to ensure that constant and even pressure is applied to the pad during curing.

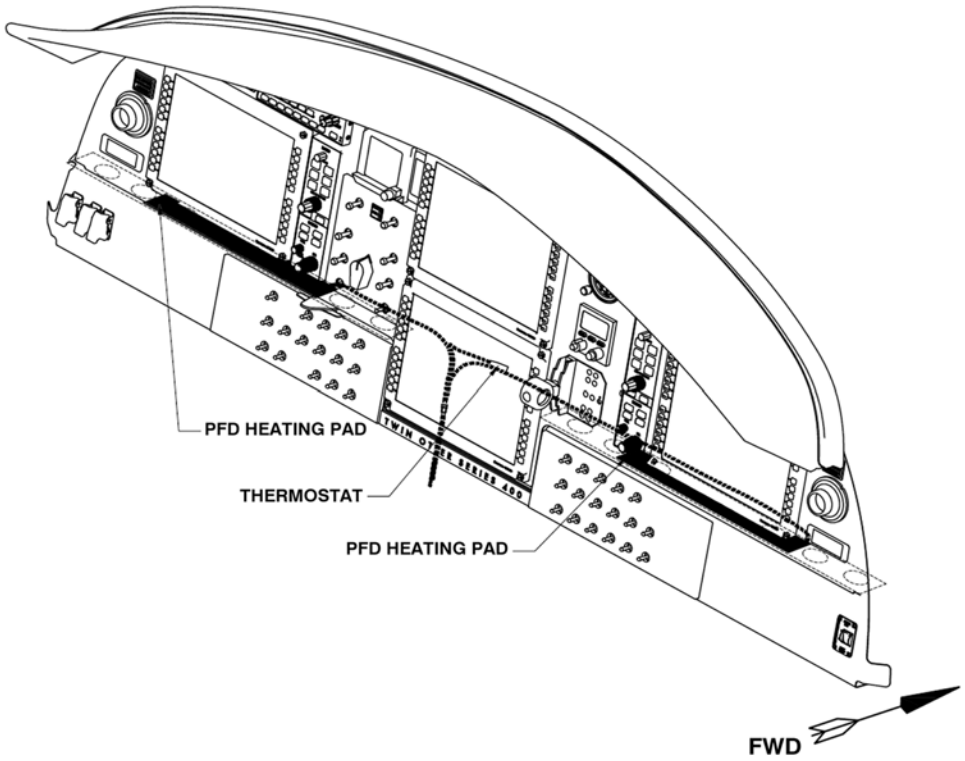
k) Once the bonding agent has cured (refer to Para 4.B) remove bag, tape and/or fixtures.

l) Apply a dollop of the adhesive/sealant under the base of the element lead (where lead enters element). This reduces wire fatigue and guards against the incursion of fluids at this location.

m) Seal the perimeter of the element with adhesive/sealant. Use just enough of the sealant to create a smooth transition along the edge of the element to mounting surface. This transition is not to extend beyond one quarter of an inch (.25") from the element and not more than one eighth of an inch (0.125") onto the elements upper surface.

n) Connect the heating pad electrical connector to the appropriate mating wire harness connector. Route wires to match those previously removed in Para 6.A.4).

o) Reinstall the pilot and co-pilot's PFDs, refer to 31-11-00, Primary Flight Display (PFD), Para 4.




Instrument Panel Frame
Figure 5

C. Ground Power Receptacle

1) Removal of Ground Power Receptacle

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Locate and open the left access panel to the hydraulic reservoir. (Refer to 12-00-00, Servicing – General Data).

c) Disconnect the wires from the fuse holders and power receptacle.

d) Remove the nuts that secure the fuse holders above the power receptacle and remove the fuse holders.

e) Remove the two nuts, four washers, and two screws that secure the power receptacle to the door.

f) Remove the power receptacle from the door.

2) Installation of Ground Power Receptacle

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Locate and open the left access panel to the hydraulic reservoir. (Refer to 12-00-00, Servicing – General Data).

c) Position the power receptacle into the door and secure the receptacle in place with two screws, four washers, and two nuts.

d) Position the fuse holders in place above the power receptacle. Secure the fuses in place with the provided nut.

e) Refer to PSM 1-64-2W, 25-70-00, Cold Weather Operations Kit (S.O.O.s 6287 or 6276) for wire routing and terminations. If reinstalling a power receptacle, route wires to match those previously removed in Para 6.C.1).

D. Cockpit Power Outlet

1) Removal of the Cockpit Power Outlet

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Locate the power outlet under the pilot's seat from inside the hydraulic bay. (Refer to 12-00-00, Servicing – General Data).

c) Disconnect the wires from the power receptacle.

d) Remove the large nut that secures the outlet in place.

e) Remove the power outlet.

2) Installation of the Cockpit Power Outlet

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Locate the installation location under the pilot's seat from inside the hydraulic bay. (Refer to 12-00-00, Servicing – General Data).

c) Position the power outlet in the channel and secure it in place with the Teflon washer and large nut provided with the outlet.

d) Refer to PSM 1-64-2W, 25-70-00, Cold Weather Operations kit (S.O.O.s 6287 or 6276) for wire routing and terminations. If reinstalling the outlet, route wires to match those previously removed in Para 6.D.1).

E. Indicator Lights

1) Removal of the Indicator Lights

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Locate the indicator lights on the left side of the aircraft.

c) Remove the electrical connector from the end of the indicator light leads by freeing the pins.

d) Remove the indicator lights by pressing in the tabs on the sides of the light's body to release the light from its location.

Note The light has been sealed in place and may require careful persuasion to free it from its boots.

2) Installation of the Indicator Lights

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Locate the installation location on the left side of the aircraft.

c) Position the indicator lights into the installation hole. Place a small amount of appropriate sealant around the head of the indicator light, behind the chrome ring. Snap the light into place by pushing it into the installation hole. Wipe off any excess sealant.

d) Install the electrical connector to the end of the indicator light leads. Connect the indicator light connector to the appropriate mating wire harness connector.

7. Adjustment/Test

A. Battery/Avionics Heat Test

1) Locate the two thermostats and disconnect from the heating pads. One is behind the instrument panel, refer to 31-11-00, Multifunction Display (MFD), Para 4, to gain access. The other is in the battery bay, refer to 12-00-00, Servicing – General Data, Figure 2 – Servicing Diagram for the location.

2) Working on one thermostat at a time, cool it below 32°F (0°C) using an ice pack, circuit cooler spray, or other appropriate means. Check the resistance across the two power contacts with an ohmmeter, the circuit should be closed. Do the same test for the other thermostat.

3) With only one thermostat connected, warm it to above 32°F (0°C) using an appropriate means. Check the resistance across the two power contacts with an ohmmeter, circuit should be open. Do the same test for the other thermostat.

4) If the thermostats do not meet either condition outlined in steps 2) and 3) above, then they need to be replaced.

B. Functional System Test

1) Cool both thermostats below 32°F (0°C) using an ice pack, circuit cooler spray, or other appropriate means.

2) Plug the system into appropriate voltage power source, 115 VAC (S.O.O. 6287) or 230 VAC (S.O.O. 6276).

3) Wait 5 to 10 minutes then check the areas next to the heating pads, the areas should feel warm to the touch.

**WARNING POWERED HEATING PADS CAN BURN BARE SKIN.**

8. Inspection/Check

It is recommended that the system operational status be checked when any of the following events occurs:

- Hard landing
- Lighting strike
- Water immersion
- Removal/replacement of a heated part or component

A. Heating Pad System Inspection

1) Clean as required before the inspection.

2) Inspect the wiring for secure attachments by following the cables to each heating pad.

3) Examine the power receptacle, power outlet, cable leads, and junctions. If any portion of the wiring shows signs of fatigue, chafing, flexing, airflow, or vibration, then re-secure and repair as required.

4) Inspect the heating pads for signs of de-bonding. If any portion of the pad has become loose, then it may be re-bonded, refer to Para 6.

5) Inspect heating pads for areas developing gray or yellow. These areas indicate that a pad has failed or is in the process of failing due to poor contact with the adhesive. Heating pads showing signs of gray/yellow need to be replaced, refer to Para 6.

6) Do a function test as per Para 7.B.

9. Troubleshooting

The system has been wired in parallel. When the system's power is disconnected an ohmmeter can be used to check and troubleshoot functional concerns.

Note Before proceeding, ensure the aircraft electrical system is OFF and no external power supplies are connected. Disconnect the main battery quick disconnect and check that all of the heating pads are properly connected.

A. Cold Weather Operations System

1) Verify engine to airframe bonding is as per 71-00-00, Power Plant (Mod 6/2012) – General Data.

2) Verify heating system grounding by checking for continuity between the power plug ground pin, engine, and airframe.

3) Verify that there is no continuity between the power contacts and the airframe ground.

4) Measure the resistance between the power pin with an ohmmeter and record the value. Compare the value with the total system readout obtained at the time of initial installation.

5) Connect the system to an appropriate 115 VAC (S.O.O. 6287) or 230 VAC (S.O.O. 6276) voltage power source.

6) Verify that the indicator light is illuminated.

7) While the system is warming up, freeze the battery and avionics thermostat and check the heating pads for warmth. The heating pads have not warmed up significantly at this point and can be touched with bare skin, but use caution.

8) Wait 5 to 10 minutes. Check the areas next to the heating pads, the area should feel warm.

**WARNING POWERED HEATING PADS CAN BURN BARE SKIN.**

Heating pads can be tested individually with an ohmmeter by disconnecting the pad from the wiring harness and measuring the resistance across the pad's leads. Resistance values for heating pads are below.

**Table 2 115V System Values (±7%)**

| Description | Quantity | Wattage | Ohms |
|---|---|---|---|
| Battery Heating Belt | 1 | 74 | 178.7 |
| AGB Heating Pad | 2 | 120 | 110.2 |
| Oil Cooler Heating Pad | 2 | 40 | 330.6 |
| PRGB Heating Pad | 2 | 240 | 55.1 |
| AGB Heating Pad | 4 | 90 | 146.9 |
| PFD Heating Pad | 2 | 40 | 330.6 |
| Emergency Battery Heating Pad | 1 | 4 | 3306.3 |
| **Total 115V System** | **1318 Watts, 11.6 Amps, 10.7 to 9.2 Ohms** | | |

**Table 3 230V System Values (±7%)**

| Description | Quantity | Wattage | Ohms |
|---|---|---|---|
| Battery Heating Belt | 1 | 74 | 714.9 |
| AGB Heating Pad | 2 | 120 | 440.8 |
| Oil Cooler Heating Pad | 2 | 40 | 1322.5 |
| PRGB Heating Pad | 2 | 240 | 220.4 |
| AGB Heating Pad | 4 | 90 | 587.8 |
| PFD Heating Pad | 2 | 40 | 1322.5 |
| Emergency Battery Heating Pad | 1 | 4 | 13225.0 |
| **Total 230V System** | **1318 Watts, 5.7 Amps, 42.5 to 36.9 Ohms** | | |

Additional troubleshooting information is provided in Table 4 — Troubleshooting.

**Table 4 Troubleshooting**

| Trouble | Possible Cause | Correction |
|---|---|---|
| Indicator light does not illuminate when power is connected. | Circuit protection is “blown” | Troubleshoot AC power source or replace the fuse |
| | Voltage too far out of range | Troubleshoot AC power source or replace the fuse |
| System does not produce heat | Circuit protection is “blown” | Troubleshoot AC power source or replace the fuse |
| | Ground power cord is not providing power | Connect cord and check power at source (wall). |
| | Wire is broken | Check connections and wire junctions. |
| | Ground power plug is damaged | Repair and/or replace plug. |
| Some of the heating pads are not warming up | Bad heating pad(s) | Check heating pad with ohmmeter, refer to Table 2 — 115V System Values (±7%) and Table 3 — 230V System Values (±7%) for resistance values. |
| | Voltage is too far out of range | Connect to a better AC power source. |
| | Wire is broken | Check connections and wire to the heating pad. |
| Smoke or odor occurs from a newly installed system | Off gassing occurs normally on newly installed systems | Check the system for proper installation. |
| Smoke or odor occurs from system that has been installed for one month or longer | Heating pad is failing (yellow/gray areas appearing on a pad) | Refer to Para 6. for pad replacement. Inspect the rest of the system for proper installation. |
| | Heat element is dirty | Refer to Para 6. for pad removal. Clean the pad in an appropriate method. |
| Circuit protection GFI on ground power continually trips | Damaged system (shorted wire) | Inspect the system for damage and shorts. |
| PFD and Battery heating pads do not warm up | Temperature is above freezing | None. PFD and Battery heaters are thermostatically controlled and not designed to turn on above freezing. |
| | Thermostat is broken | Check function of the thermostat, refer to Para 8. |

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